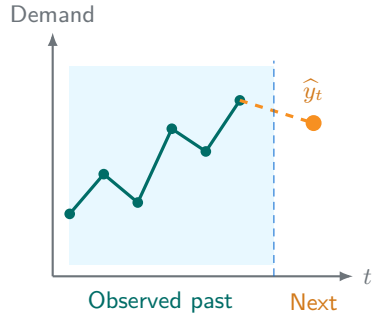
SEQUENTIAL DECISIONS WITH CONVEX LOSSES

Online Learning

From daily demand to
convex optimization

Paul Minchella
Chapters 1 and 2



The learning paradigm: model, loss and optimization I

Classical statistical setting: $(X_i, Y_i)_{i=1}^n \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{(X,Y)}$. Learn a parameterized predictor $\widehat{y} = \mu_\theta(x)$.

DEFINITION — Model and learning objective

Model class: linear models, splines, random forests, neural networks,...
Loss: measures prediction error. For squared loss,

$$\ell_{(x,y)}(\theta) = \frac{1}{2}\big(y - \mu_\theta(x)\big)^2.$$

Empirical objective:

$$L_n(\theta) = \frac{1}{n}\sum_{i=1}^n \ell_{(X_i,Y_i)}(\theta).$$

Learning: approximately minimize L_n.

DEFINITION — Gradient descent, when applicable

Gradient: direction of steepest local increase in Euclidean geometry.
Descent: move in the opposite direction,

$$\boldsymbol{\theta_{k+1} = \theta_k - \eta_k \nabla L_n(\theta_k).}$$

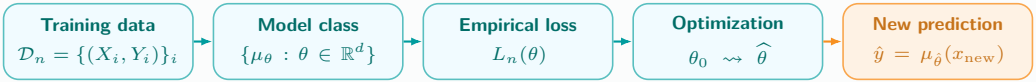
Learning rate: $\eta_k > 0$. Possible decreasing schedules:

$$\eta_k = \frac{\eta_0}{k+1} \qquad \text{or} \qquad \eta_k = \frac{\eta_0}{\sqrt{k+1}}.$$

Here, k counts optimization iterations.

The learning paradigm: model, loss and optimization II

Convergence requires suitable assumptions and step sizes. A suitable constant step can also work. Neural-network losses are generally nonconvex in their parameters; random forests use a different fitting procedure.



After K iterations: $\widehat{\theta} = \theta_K$, an approximate solution, with no automatic guarantee of global optimality.
Next question: how should prediction use observations ***arriving over time***?

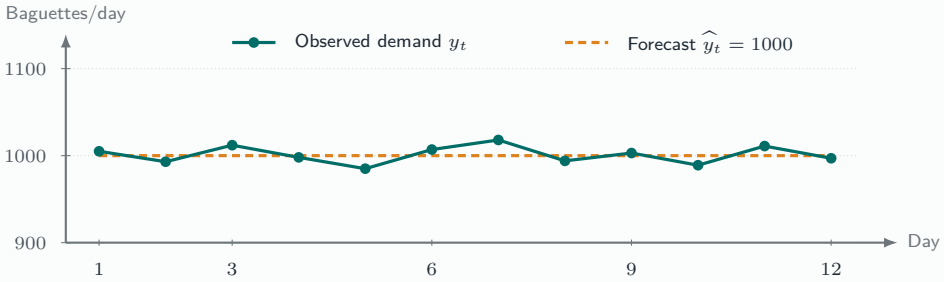
A bakery: daily demand around a stable level

EXAMPLE — A constant mean with small daily fluctuations

Observed demand:

$$y_t = 1000 + \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \text{Unif}\{-20, -19, \ldots, 20\}.$$

Prediction model: $\widehat{y}_t = \theta$. **Optimal level under squared loss:** $\theta^\star = 1000$.



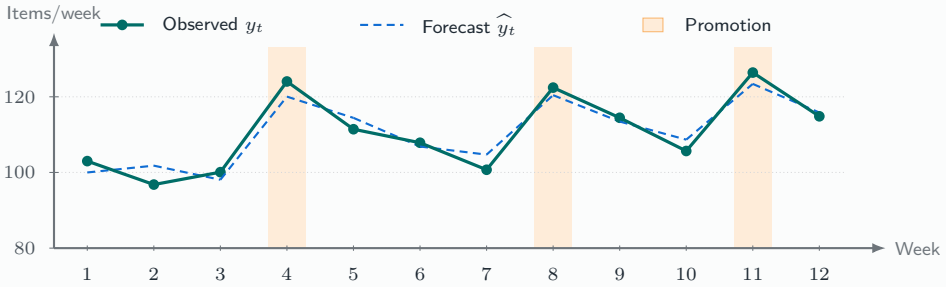
A clothing item: weekly demand with temporal dynamics

EXAMPLE — An illustrative model for a WIP item

$$y_t = \underbrace{100 + 0.6(y_{t-1} - 100) + 20x_t}_{\widehat{y}_t} + \varepsilon_t.$$

Persistence: $\rho = 0.6$.

Promotion: $x_t \in \{0, 1\}$, known before prediction.



Synthetic trajectory with $y_0 = 100$. Assume $\mathbb{E}[\varepsilon_t \mid y_{1:t-1}, x_t] = 0$. The forecast uses the model parameters shown above.

Time series and the online learning objective I

DEFINITION — Stochastic process and observed time series

A **stochastic process** is a family $X = (X_t)_{t \in \mathcal{T}}$ of random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. If $X_t \in \mathbb{R}^d$, it is a vector process.
For a fixed $\omega \in \Omega$, a **trajectory** is the map $t \longmapsto X_t(\omega)$. An **observed time series** is a finite portion of one trajectory:

$$x_1 = X_1(\omega), \ldots, x_T = X_T(\omega).$$

Course objective: learning while decisions are being made

Before observing y_t	Predict $\widehat{y}_t = \mu_{\theta_t}(x_t)$ using past feedback and covariates available at prediction time.
After observing y_t	Incur the loss $\ell_t(\theta_t)$ and obtain feedback.
Before the next round	Update θ_t to θ_{t+1} using this new information.

Time series and the online learning objective II

Temporal dependence: a property of the data.
Online learning: a protocol for sequential decisions and updates.
A time-series model can also be fitted offline on a fixed historical sample.

Statistical learning vs. robust online learning I

Shared ingredients: a predictor μ_θ, a loss function, and admissible parameters $\theta \in \mathcal{K}$.

DEFINITION — Statistical learning

Data assumption: observations sampled from an unknown, fixed distribution,

$$(X_i, Y_i)_{i=1}^n \overset{\text{i.i.d.}}{\sim} \mathbb{P}.$$

Population target: minimize expected loss,

$$\min_{\theta \in \mathcal{K}} \mathbb{E}_{(X,Y)\sim\mathbb{P}} \left[\ell_{(X,Y)}(\theta) \right].$$

Empirical fit: estimate the parameters through

$$\widehat{\theta} \in \arg\min_{\theta \in \mathcal{K}} \frac{1}{n} \sum\nolimits_{i=1}^n \ell_{(X_i,Y_i)}(\theta).$$

DEFINITION — Adversarial online learning

Sequential decision: use past feedback,

$$\theta_t = \pi_t(\ell_1, \ldots, \ell_{t-1}) \in \mathcal{K}.$$

Nature: reveals the current loss $\ell_t \in \mathcal{L}$ after the decision.

No stochastic law assumed: losses may form an adversarial sequence.

Worst-case target: choose a causal strategy against the most unfavorable admissible loss sequence,

$$\min_{\pi \in \mathfrak{P}} \max_{\ell_{1:T} \in \mathcal{L}^T} \sum\nolimits_{t=1}^T \ell_t(\theta_t),$$
$$\theta_t = \pi_t(\ell_1, \ldots, \ell_{t-1}).$$

Statistical learning vs. robust online learning II

$$\mathrm{Reg}_T(\pi; \ell_{1:T}) := \underbrace{\sum_{t=1}^{T} \ell_t(\theta_t)}_{\text{online cumulative loss}} - \underbrace{\min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u)}_{\text{best fixed parameter in hindsight}} .$$

Adversarial viewpoint: seek a regret guarantee valid for *every admissible loss sequence*, including unfavorable responses from Nature.

$\mathfrak{P}$: causal decision rules. For the OCO guarantees studied here, $\mathcal{K}$ is bounded, closed and convex, and $\mathcal{L}$ contains convex losses with a common Lipschitz bound.

Contents

1. Motivation: why sequential learning?

2. Online protocol and the first mathematical definitions

3. How performance is measured: regret

4. Online algorithms and regret guarantees

5. Applications: regression and inventory

Learning objective
Formulate a sequential prediction problem and understand the convex geometry used to analyse online updates.

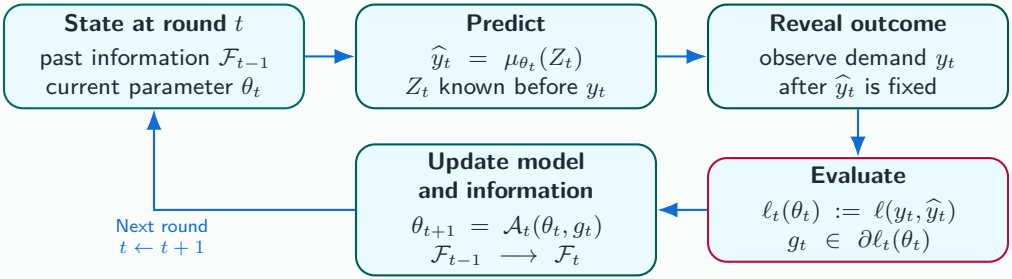
CHAPTER 1

Motivation

Why must a prediction be made before the data arrive?

One product, one store, one prediction each morning

The store predicts today's demand **before sales reveal it**. New observations then improve the next decision, without requiring a full batch refit.



WHY SEQUENTIALIZE?

Data arrive over time, decisions cannot wait, and demand may change. The update uses feedback **only after** the current prediction is fixed.

Reminder: gradient descent for one fixed loss

DEFINITION — Gradient descent

For a differentiable loss $\ell : \mathbb{R}^d \to \mathbb{R}$, a learning rate $\eta > 0$ gives $\boldsymbol{\theta_{t+1} = \theta_t - \eta \nabla \ell(\theta_t)}$. The gradient gives a local slope. The learning rate controls the size of the correction.

Reminder: gradient descent for one fixed loss

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THEOREM — Convergence in the stationary strongly convex case

Assume that ℓ is L-smooth and μ-strongly convex on $\mathbb{R}^d$, with $0 < \mu \leq L$. It has a unique minimizer $\theta^\star$. For a constant step $0 < \eta \leq 1/L$,

$$\ell(\theta_t) - \ell(\theta^\star) \leq (1 - \mu\eta)^{t-1} \big[\ell(\theta_1) - \ell(\theta^\star)\big].$$

Moreover, $\frac{\mu}{2}\|\theta_t - \theta^\star\|_2^2 \leq \ell(\theta_t) - \ell(\theta^\star)$. Hence $\theta_t \xrightarrow[t\to\infty]{} \theta^\star$ at a geometric rate.

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This result repeatedly optimizes **the same function**. Online learning will *replace it by a sequence* $\ell_1, \ell_2, \ldots$

Constant demand: learning to predict I

The demand is always $y_t = 14$. We predict $\widehat{y}_t = \theta_t$, starting from $\theta_0 = \widehat{y}_0 = 10$.

$$\ell(\theta) = \frac{1}{2}(\theta - 14)^2, \qquad \nabla\ell(\theta) = \theta - 14.$$

With learning rate $\eta = 0.25$: $\boldsymbol{\theta_{t+1} = \theta_t - 0.25(\theta_t - 14) = 0.75\theta_t + 0.25 \times 14.}$

t	θ_t	$-\eta\nabla\ell(\theta_t)$	$\ell(\theta_t)$
0	10.000	1.000	8.000
1	11.000	0.750	4.500
2	11.750	0.563	2.531
3	12.313	0.422	1.424
4	12.734	0.316	0.801
5	13.051	0.237	0.451

Values rounded to three decimal places.

Convergence to the target

The distance to the target contracts:

$$\theta_{t+1} - 14 = 0.75(\theta_t - 14).$$

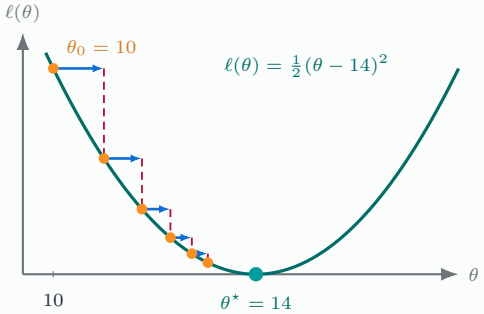
Hence, since $\theta_0 = 10$,

$$\theta_t = 14 - 4\left(\frac{3}{4}\right)^t.$$

Therefore,

$$\theta_t \xrightarrow[t\to\infty]{} 14$$

Constant demand: learning to predict II



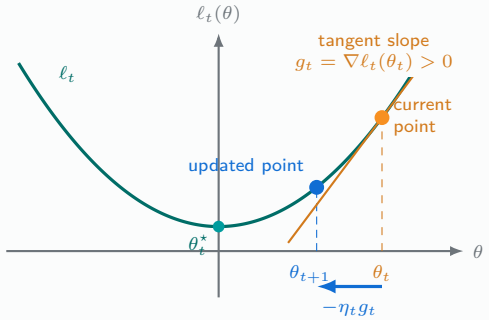
Horizontal arrows: negative-gradient steps.
Dashed segments: decrease in the loss.

WHAT CHANGES WHEN DEMAND VARIES?

With varying demand y_t, the loss becomes $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$. Its minimizer moves: the aim is now to **adapt to changing demand**, using each observation to prepare the next prediction.

From a fixed loss to a sequence of losses

A new loss ℓ_t may have a new minimizer $\theta_t^\star$. Each observation updates the learner's parameter.



DEFINITION — One online gradient step

$$\theta_{t+1} = \theta_t - \eta_t \nabla \ell_t(\theta_t).$$

The slope changes the *next* decision.

EXAMPLE — The illustrated step

$\ell_t(\theta) = \frac{1}{2}\theta^2 + 0.45$, $\theta_t = 2$ and $\eta_t = 0.4$ give

$$\theta_{t+1} = 1.2, \qquad 1.17 < 2.45.$$

Strict decrease holds for L-smooth ℓ_t, $g_t \neq 0$ and $0 < \eta_t < 2/L$.

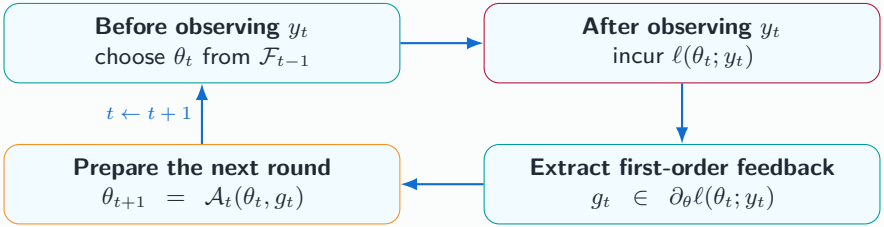
A CHANGING OBJECTIVE

A lower ℓ_t does not ensure a lower ℓ_{t+1}. *Tracking* a moving optimum needs additional assumptions.

The same loss formula, new data at every round

Even with a fixed analytical form, a new observation defines a new loss:

$$\ell_t(\theta) = \ell(\theta; y_t).$$



EXAMPLE — From a scalar forecast to a parametric model

With features $x_t \in \mathbb{R}^d$ known before y_t,

$$\widehat{y}_t = x_t^\top \theta_t, \qquad \ell_t(\theta) = \tfrac{1}{2}(x_t^\top \theta - y_t)^2, \qquad g_t = (x_t^\top \theta_t - y_t) x_t.$$

Setting $x_t \equiv 1$ recovers a scalar forecast. Features may encode promotions or calendar effects.

CHAPTER 2

Online protocol and mathematical definitions

Feasible decisions, supporting slopes and the assumptions behind learning guarantees

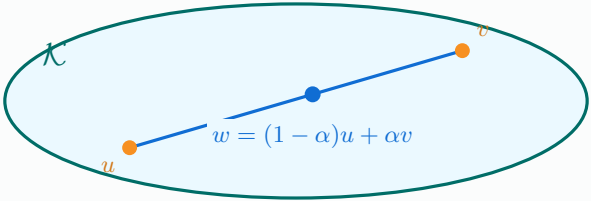
Convex sets: feasible decisions

DEFINITION — Convex set

A set $\mathcal{K} \subseteq \mathbb{R}^d$ is **convex** if

$$\forall\, u, v \in \mathcal{K}, \quad \forall\, \alpha \in [0,1], \qquad \boldsymbol{(1-\alpha)u + \alpha v \in \mathcal{K}.}$$

The entire line segment joining any two points of $\mathcal{K}$ remains inside $\mathcal{K}$.



Interpretation: interpolating between two admissible decisions produces another admissible decision.

In the online convex optimization setting studied here, $\mathcal{K}$ is also assumed nonempty and closed.

Convex functions: the graph lies below its chords

DEFINITION — Convex function

Let $\mathcal{K} \subseteq \mathbb{R}^d$ be convex. A function $f : \mathcal{K} \to \mathbb{R}$ is **convex** if, for all $u, v \in \mathcal{K}$ and $\alpha \in [0, 1]$,

$$f\big((1-\alpha)u + \alpha v\big) \leq (1-\alpha) f(u) + \alpha f(v).$$

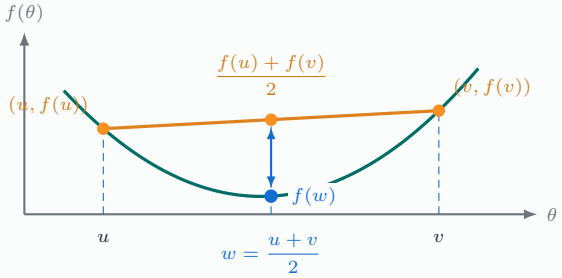


Illustration with $\alpha = \frac{1}{2}$: the blue point lies below the orange chord. Convexity requires this inequality for every pair of points and every $\alpha \in [0, 1]$.

Subgradients: an affine lower bound I

DEFINITION — Subgradient relative to $\mathcal{K}$

For convex $f : \mathcal{K} \to \mathbb{R}$, a subgradient $g \in \mathbb{R}^d$ at u satisfies

$$\forall v \in \mathcal{K}, \qquad f(v) \geq f(u) + \langle g, v - u \rangle.$$

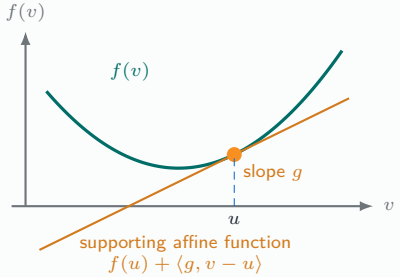
The set of these vectors is $\partial_{\mathcal{K}} f(u)$. For $\mathcal{K} = \mathbb{R}^d$, we simply write $\partial f(u)$.

At a differentiable **interior point**,

$$\partial_{\mathcal{K}} f(u) = \{\nabla f(u)\}.$$

The relative subdifferential may be larger at the boundary.

Subgradients: an affine lower bound II



PROPOSITION — Convex linearization inequality

$$f(u) - f(v) \leq \langle g, u - v \rangle, \qquad g \in \partial_{\mathcal{K}} f(u).$$

Why it matters. Set $u = \theta_t$ and $v = \theta$: a linear expression bounds $\ell_t(\theta_t) - \ell_t(\theta)$, even for nondifferentiable losses.

From absolute loss to asymmetric pinball loss I

EXAMPLE — The symmetric starting point

For $\ell_{\mathrm{abs}}(\theta; y) = |y - \theta|$, the slope is -1 if $\theta < y$ and $+1$ if $\theta > y$. At the kink, $\partial_\theta \ell_{\mathrm{abs}}(y; y) = [-1, 1]$.

DEFINITION — Pinball loss, $0 < \tau < 1$

$\ell(\theta; y) = \rho_\tau(y - \theta)$, where

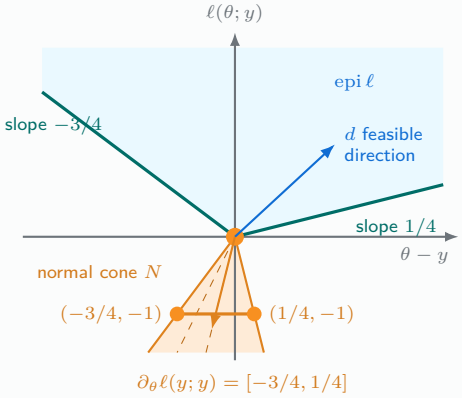
$$\rho_\tau(r) = \begin{cases} \tau r, & r \geq 0, \\ (\tau - 1) r, & r < 0. \end{cases}$$

$$\partial_\theta \ell(\theta; y) = \begin{cases} \{-\tau\}, & \theta < y, \\ [-\tau, 1 - \tau], & \theta = y, \\ \{1 - \tau\}, & \theta > y. \end{cases}$$

$\rho_{1/2}(y - \theta) = \frac{1}{2}|y - \theta|$ recovers the symmetric case.

With $\tau = 3/4$: underprediction gives $+3\eta_t/4$, overprediction gives $-\eta_t/4$.

From absolute loss to asymmetric pinball loss II



Normals satisfy $\langle z, d \rangle \leq 0$ for every feasible direction d. The slice $(g, -1) \in N$ gives the subgradients.

Unequal shortage and surplus costs will determine τ. References: Koenker and Bassett (1978), Gneiting (2011).

Subgradients and first-order optimality I

PROPOSITION — Optimality relative to $\mathcal{K}$

For convex $f : \mathcal{K} \to \mathbb{R}$,

$$x^\star \in \arg\min_{x \in \mathcal{K}} f(x) \quad \Longleftrightarrow \quad 0 \in \partial_{\mathcal{K}} f(x^\star).$$

Proof. Both mean $f(v) \geq f(x^\star)$ for all $v \in \mathcal{K}$.

At a differentiable interior minimizer, $\nabla f(x^\star) = 0$.

DEFINITION — The one-dimensional form

For finite convex $f : I \to \mathbb{R}$ and $x \in \operatorname{int} I$,

$$\partial_I f(x) = [f'_-(x), f'_+(x)].$$

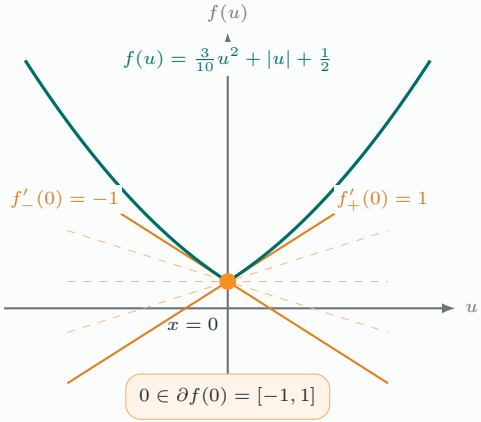
Optimality means $f'_-(x) \leq 0 \leq f'_+(x)$.

Convexity orders the secant slopes:

$$f'_-(x) = \sup_{v \in I,\, v < x} \frac{f(v) - f(x)}{v - x}, \qquad f'_+(x) = \inf_{v \in I,\, v > x} \frac{f(v) - f(x)}{v - x}.$$

Subgradients and first-order optimality II

At the boundary: $f(x) = x$ on $[0, \infty)$ is minimized at 0, yet $f'(0) = 1$. The relative subdifferential $(-\infty, 1]$ contains 0.



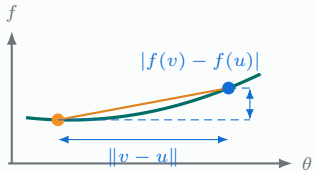
Three regularity notions, three different roles

G-Lipschitz

Function values vary at most linearly with distance:

$$|f(v) - f(u)| \le G\|v - u\|.$$

Controls the magnitude of first-order changes.

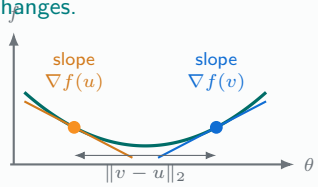


β-smooth

A differentiable f has a Lipschitz gradient:

$$\|\nabla f(v) - \nabla f(u)\|_2 \le \beta\|v - u\|_2.$$

Controls how quickly the slope changes.

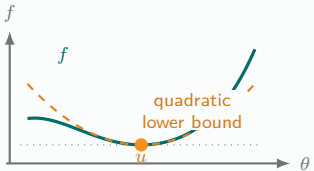


α-strongly convex

For $\alpha > 0$ and every $g \in \partial_{\mathcal{K}} f(u)$,

$$f(v) \ge f(u) + \langle g, v - u\rangle + \tfrac{\alpha}{2}\|v - u\|_2^2.$$

Supplies a quadratic lower bound in every direction.



affine support at u

If $\|\nabla f(u)\|_2 \le G$ throughout a convex domain, then f is G-Lipschitz there.

Example: $f(\theta) = \frac{1}{2}\theta^2$ is 1-smooth and 1-strongly convex on $\mathbb{R}$, but it is not globally Lipschitz.

Asymptotic notation and sublinear growth

Let (a_T) be real-valued and let $b_T > 0$ for all sufficiently large T.

DEFINITION — Big-O and little-o

- $a_T = O(b_T)$ means that some $C > 0$ and T_0 satisfy
$$|a_T| \le C b_T \qquad \text{for all } T \ge T_0.$$
- $a_T = o(b_T)$ means $|a_T|/b_T \to 0$ as $T \to \infty$.

DEFINITION — Sublinear growth

$$\boldsymbol{a_T = o(T) \iff \frac{|a_T|}{T} \longrightarrow 0.}$$

No bounded but no regret

From cumulative to average scale

Cumulative bound	Average bound
$O(1)$	$O(1/T)$
$O(\log T)$	$O(\log T/T)$
$O(\sqrt{T})$	$O(1/\sqrt{T})$
$O(T^\alpha),\ 0 \le \alpha < 1$	$O(T^{\alpha-1})$

$$|a_T| \le C\sqrt{T} \quad \Longrightarrow \quad \frac{|a_T|}{T} \le \frac{C}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0.$$

Sublinear does not mean bounded: $\sqrt{T}$ and $\log T$ diverge, while their averages vanish.

A sublinear regret upper bound makes the possible positive average excess loss **vanish**. It does not imply parameter convergence.

CHAPTER 3

Regret

A cumulative benchmark for decisions made without knowing the future

The benchmark: one fixed decision chosen in hindsight

A NATURAL BENCHMARK

The best fixed decision after observing the entire sequence.

DEFINITION — The learner: sequential decisions

Information: past observations only.
Decisions: $\theta_1, \dots, \theta_T \in \mathcal{K}$.
Cumulative loss:

$$L_T^{\mathrm{alg}} = \sum_{t=1}^{T} \ell_t(\theta_t).$$

DEFINITION — The oracle: a hindsight comparator

Information: all losses $\ell_1, \dots, \ell_T$.
Decision: one fixed parameter,

$$\theta_T^\star \in \arg\min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u).$$

Same losses and feasible set. One fixed oracle decision for all rounds.
Regret: cumulative learner loss minus cumulative benchmark loss.

Assume the minimum is attained. Otherwise, define the benchmark using an infimum.

Static regret and the no-regret objective

DEFINITION — Static regret

Fixed comparator $u \in \mathcal{K}$; best fixed comparator:

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} \big[\ell_t(\theta_t) - \ell_t(u)\big],$$

$$\mathrm{Reg}_T = \sum_{t=1}^{T} \ell_t(\theta_t) - \min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u) = L_T^{\mathrm{alg}} - L_T^{\star}.$$

DEFINITION — No-regret: a sublinear upper bound

Uniform guarantee: $\mathrm{Reg}_T \le B_T$ for every admissible loss sequence, with $B_T \ge 0$ and $B_T = o(T)$.

$$\frac{1}{T}\sum_{t=1}^{T} \ell_t(\theta_t) \le \min_{u \in \mathcal{K}} \frac{1}{T}\sum_{t=1}^{T} \ell_t(u) + \frac{B_T}{T}, \qquad \frac{B_T}{T} \longrightarrow 0.$$

Regret can be negative. The guarantee controls $\max\{\mathrm{Reg}_T, 0\}/T \to 0$. When regret is nonnegative, this is $\mathrm{Reg}_T/T \to 0$, equivalently $\mathrm{Reg}_T = o(T)$.

Average performance only: no pointwise accuracy, parameter convergence or tracking guarantee.

Running example: four demands, one hindsight benchmark

Setup: $\mathcal{K} = [0, 20]$, $\widehat{y}_t = \theta_t$, $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$.
Update: $\theta_{t+1} = \theta_t - 0.25(\theta_t - y_t)$. $\quad \theta_1 = 10$, $\eta = 0.25$.

t	θ_t **chosen**	y_t **revealed**	$g_t = \theta_t - y_t$	$\ell_t(\theta_t)$
1	10.000	12	-2.000	2.000
2	10.500	15	-4.500	10.125
3	11.625	11	0.625	0.195
4	11.469	18	-6.531	21.329

EXAMPLE — The hindsight calculation

Best constant prediction: $\theta_4^\star = \frac{1}{4}(12 + 15 + 11 + 18) = 14$.

$$L_4^{\text{alg}} \approx 33.649, \qquad L_4^\star = \tfrac{1}{2}(4 + 1 + 9 + 16) = 15.$$

$$\mathbf{Reg_4 = L_4^{alg} - L_4^\star \approx 18.65.}$$

Exact values before rounding; all updates remain in $\mathcal{K}$.

Four rounds and initial offset: a large finite regret can coexist with a sublinear bound.

$\eta = 0.25$ is illustrative. A no-regret guarantee needs suitable assumptions and step-size choices.

From a performance criterion to online algorithms

Regret defines the objective. Compete with the best fixed hindsight decision.
An algorithm defines the updates. Use past feedback to choose the next decision.

Method	Update principle	Central question
FTL	Minimize past losses	Stability of past-loss minimizers
FTRL	Add regularization	Stability through regularization
OGD	One (sub)gradient step	Local corrections, cumulative control

TARGET BOUNDS UNDER SUITABLE ASSUMPTIONS AND TUNING

$$\mathrm{Reg}_T \le B_T, \qquad \boldsymbol{B_T/T \longrightarrow 0.}$$

Typical scales: $O(\sqrt{T})$ for convex losses; $O(\log T)$ with additional strong convexity.
Proof ingredients: gradient bounds, feasible-set geometry, learning rates.

FTL may be unstable. The following chapter studies how regularization, geometry and step-size choices turn sequential updates into regret guarantees.

CHAPTER 4

Online algorithms

Stability, regularization and first-order updates

Adaptation and stability I

THE ALGORITHMIC CHALLENGE

Adaptation: react to newly observed losses.
Stability: limit sensitivity to a single observation.
Information: past feedback only, no access to future losses.

Common setting: a nonempty closed convex set $\mathcal{K} \subseteq \mathbb{R}^d$, convex losses ℓ_t and decisions $\theta_t \in \mathcal{K}$. Assume the minimizers used below exist.

DEFINITION — First-order feedback at the current decision

$$g_t \in \partial \ell_t(\theta_t), \qquad \ell_t(u) \geq \ell_t(\theta_t) + \langle g_t, u - \theta_t \rangle \quad (u \in \mathcal{K}).$$

Differentiable loss: use $g_t = \nabla \ell_t(\theta_t)$.
Nonsmooth loss: choose a valid subgradient.

Adaptation and stability II

Convexity turns loss differences into linear comparisons:

$$\ell_t(\theta_t) - \ell_t(u) \leq \langle g_t, \theta_t - u \rangle.$$

At a constrained boundary, the relative subdifferential can contain more than the gradient. Gradient bounds below concern the selected g_t.

Follow the Leader (FTL) I

DEFINITION — The empirical leader

Initialization: any $\theta_1 \in \mathcal{K}$.
After observing ℓ_t: minimize all past losses,

$$F_t(u) := \sum_{s=1}^{t} \ell_s(u), \qquad \boldsymbol{\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} F_t(u).}$$

Intuition: play tomorrow the decision that would have worked best so far.

Feedback: past loss functions, or sufficient statistics.

Limitation: a small change in the objective can switch the leader.

Follow the Leader (FTL) II

EXAMPLE — Squared loss: the leader is a running average

For $\ell_s(u) = \frac{1}{2}(u - y_s)^2$ and $\mathcal{K} = \mathbb{R}$:

$$\theta_{t+1} = \frac{1}{t} \sum_{s=1}^{t} y_s.$$

On $\mathcal{K} = [a, b]$, project this average onto $[a, b]$.

Convexity alone does not guarantee sublinear FTL regret. For linear losses on a bounded interval, leaders can oscillate between endpoints.

FTL analysis: the cost of delaying the leader I

PROPOSITION — Be-the-Leader inequality

For every $u \in \mathcal{K}$,

$$\sum_{t=1}^{T} \ell_t(\theta_{t+1}) \leq \sum_{t=1}^{T} \ell_t(u).$$

Analysis device: θ_{t+1} already uses the current loss ℓ_t.

Regret decomposition:

$$\mathrm{Reg}_T(u) = \underbrace{\sum_{t=1}^{T} [\ell_t(\theta_t) - \ell_t(\theta_{t+1})]}_{\text{cost of the one-round delay}} + \underbrace{\sum_{t=1}^{T} [\ell_t(\theta_{t+1}) - \ell_t(u)]}_{\leq 0}.$$

FTL analysis: the cost of delaying the leader II

PROPOSITION — FTL stability bound

$$\mathrm{Reg}_T(u) \leq \sum_{t=1}^{T} [\ell_t(\theta_t) - \ell_t(\theta_{t+1})].$$

Proof target: control the total change in loss when the leader updates.
Stable leaders: small cumulative cost of acting one round earlier.

FTL with strongly convex losses I

Assumptions: every ℓ_t is α-strongly convex, $\alpha > 0$, and the selected $g_t \in \partial \ell_t(\theta_t)$ satisfy $\|g_t\|_2 \le G$.

THEOREM — Logarithmic regret

For every $u \in \mathcal{K}$,

$$\mathbf{Reg}_T(u) \le \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t} \le \frac{G^2}{2\alpha}(1 + \log T).$$

Why the harmonic sum appears

Accumulated curvature: $F_t = \sum_{s=1}^{t} \ell_s$ is $t\alpha$-strongly convex.

Set $\Delta_t := F_t(\theta_t) - F_t(\theta_{t+1})$. Then

$$\mathrm{Reg}_T = \sum_{t=1}^{T} \Delta_t, \qquad \Delta_t \le G\|\theta_t - \theta_{t+1}\|_2 - \frac{\alpha t}{2}\|\theta_t - \theta_{t+1}\|_2^2 \le \frac{G^2}{2\alpha t}.$$

FTL with strongly convex losses II

CURVATURE SUPPLIES STABILITY

More observations increase the curvature of the cumulative objective.
Per-round bound: $1/t$. **Cumulative bound:** $1 + \log T$.

The identity uses $F_0 \equiv 0$ and exact FTL minimizers. The gradient bound remains essential, even with quadratic losses.

Follow the Regularized Leader (FTRL) I

ADDING CURVATURE WHEN LOSSES DO NOT PROVIDE IT

A strongly convex regularizer R stabilizes the cumulative objective.
Its weight $1/\eta$ controls the influence of this regularization.

DEFINITION — The regularized leader

Input: a strongly convex $R : \mathcal{K} \to \mathbb{R}$ and $\eta > 0$.
Initialization: $\theta_1 \in \arg\min_{u \in \mathcal{K}} R(u)$.
After observing ℓ_t**:**

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \left\{ \sum_{s=1}^{t} \ell_s(u) + \frac{1}{\eta} R(u) \right\}.$$

Follow the Regularized Leader (FTRL) II

Smaller η	Stronger regularization, greater pull toward the minimizer of R
Larger η	Weaker regularization, more influence from observed losses
Choice of R	Geometry and stability of the updates

The next slides analyze FTRL on linearized losses. Curvature of R and curvature of the original losses are separate assumptions.

Linearized FTRL: the quadratic case I

Affine lower model: $\ell_s(\theta_s) + \langle g_s, u - \theta_s \rangle \leq \ell_s(u)$.

Constants in u: no effect on the minimizer.

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \left\{ \left\langle \sum_{s=1}^{t} g_s,\, u \right\rangle + \frac{1}{\eta} R(u) \right\}.$$

EXAMPLE — Unconstrained domain and quadratic regularizer

$\mathcal{K} = \mathbb{R}^d$, $R(u) = \frac{1}{2}\|u\|_2^2$, $\eta > 0$, hence $\theta_1 = 0$.

$$J(u) = \left\langle \sum_{s=1}^{t} g_s,\, u \right\rangle + \frac{1}{2\eta}\|u\|_2^2, \qquad \nabla_u J(u) = \sum_{s=1}^{t} g_s + \frac{1}{\eta} u.$$

$$\nabla_u^2 J(u) = \frac{1}{\eta} I_d \succ 0, \qquad \nabla_u J(\theta_{t+1}) = 0 \quad \Longrightarrow \quad \boldsymbol{\theta_{t+1} = -\eta \sum_{s=1}^{t} g_s.}$$

Linearized FTRL: the quadratic case II

Fixed η, no constraints: $\theta_{t+1} = \theta_t - \eta g_t$. The iterates coincide with OGD initialized at 0.

With constraints:

$$\theta_{t+1} = \Pi_{\mathcal{K}} \left(-\eta \sum_{s=1}^{t} g_s \right),$$

generally different from repeated projected OGD steps.

The vectors g_s are fixed when differentiating J with respect to u. Quadratic regularization ensures a unique finite minimizer at each round.

Linearized FTRL: a square-root regret bound I

Assumptions: convex losses, $\|g_t\|_2 \le G$, and an α_R-strongly convex regularizer R in Euclidean norm.

$$D_R := \sup_{u \in \mathcal{K}} \big[R(u) - R(\theta_1)\big] < \infty, \qquad \theta_1 \in \arg\min_{v \in \mathcal{K}} R(v).$$

THEOREM — Regularization cost and stability cost

$$\mathrm{Reg}_T(u) \le \frac{R(u) - R(\theta_1)}{\eta} + \frac{\eta}{2\alpha_R} \sum_{t=1}^{T} \|g_t\|_2^2.$$

In particular,

$$\mathrm{Reg}_T \le \underbrace{\frac{D_R}{\eta}}_{\text{regularization cost}} + \underbrace{\frac{\eta G^2 T}{2\alpha_R}}_{\text{stability cost}}.$$

Linearized FTRL: a square-root regret bound II

Known horizon T, $D_R, G > 0$: balance the two terms,

$$\eta = \frac{\sqrt{2\alpha_R D_R}}{G\sqrt{T}} \quad \Longrightarrow \quad \mathrm{Reg}_T \le G\sqrt{\frac{2D_R T}{\alpha_R}}, \qquad \frac{\mathrm{Reg}_T}{T} \le G\sqrt{\frac{2D_R}{\alpha_R T}}.$$

α_R measures regularizer curvature. On $\mathbb{R}^d$ with quadratic R, $D_R = \infty$: use the comparator-dependent first inequality instead of this uniform bound.

Vanishing average regret

DEFINITION — An upper bound on average excess loss

If $\mathrm{Reg}_T \le B_T$ and $B_T/T \to 0$, then

$$\frac{1}{T}\sum_{t=1}^{T} \ell_t(\theta_t) \le \min_{u \in \mathcal{K}} \frac{1}{T}\sum_{t=1}^{T} \ell_t(u) + \frac{B_T}{T}.$$

Method and assumptions	B_T	B_T/T
FTL, strongly convex losses	$O(\log T)$	$O(\log T/T)$
Linearized FTRL, convex losses	$O(\sqrt{T})$	$O(T^{-1/2})$

Benchmark: one fixed decision selected with hindsight.
Guarantee: asymptotically no larger average loss, up to a vanishing term.

Scope: cumulative performance. Parameter convergence, per-round optimality and tracking a moving oracle require additional analysis.

The previous assumptions and tuning apply. Regret can be negative: these rates describe upper bounds, not its absolute magnitude.

Projected Online Gradient Descent (OGD)

One local correction per round. Initialize $\theta_1 \in \mathcal{K}$ and choose positive learning rates η_t.

Stage	Information and action at round t
History	Past decisions and feedback determine θ_t
Decision	Play θ_t before observing the current loss
Feedback	Observe the loss incurred and a subgradient $g_t \in \partial \ell_t(\theta_t)$
Update	Store $\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t)$ for the next round

DEFINITION — Euclidean projection

$$\Pi_{\mathcal{K}}(z) := \arg\min_{u \in \mathcal{K}} \|u - z\|_2^2, \qquad \boldsymbol{\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t).}$$

Direction: $-g_t$. **Magnitude:** η_t. **Feasibility:** $\Pi_{\mathcal{K}}$.

Benefit: no re-optimization of the full loss history.

A scalar loss value alone does not generally determine g_t. Projection may be costly on a complex feasible set.

Projected OGD: two rounds in chronological order

Setup: $\mathcal{K} = [0, 20]$, $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$, $\theta_1 = 18$, $\eta_t = 0.25$.

$$g_t = \theta_t - y_t, \qquad \widetilde{\theta}_{t+1} = \theta_t - 0.25 g_t, \qquad \Pi_{[0,20]}(z) = \min\{20, \max\{0, z\}\}.$$

Chronological stage	**Round** $t = 1$	**Round** $t = 2$
1. Play θ_t	18	20
2. Reveal y_t	30	14
3. Incur $\ell_t(\theta_t)$	$\frac{1}{2}(18 - 30)^2 = 72$	$\frac{1}{2}(20 - 14)^2 = 18$
4. Compute g_t	-12	6
5. Tentative update	$18 - 0.25(-12) = 21$	$20 - 0.25(6) = 18.5$
6. Project and store	$\theta_2 = \Pi_{\mathcal{K}}(21) = 20$	$\theta_3 = \Pi_{\mathcal{K}}(18.5) = 18.5$

Round 1: prediction too low, upward correction, projection active.
Round 2: prediction too high, downward correction, projection inactive.

The feasible set constrains predictions, not observations: demand $y_1 = 30$ is allowed. Each θ_t is chosen before y_t is revealed. The constant step is illustrative.

OGD analysis: a telescoping distance bound

Assumptions: convex ℓ_t, $\|g_t\|_2 \le G$, $\operatorname{diam}(\mathcal{K}) \le D$ and constant $\eta > 0$.
Projection and convexity: for any $u \in \mathcal{K}$,

$$\ell_t(\theta_t) - \ell_t(u) \le \langle g_t, \theta_t - u \rangle \le \frac{\|\theta_t - u\|_2^2 - \|\theta_{t+1} - u\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

Projection cannot increase the distance to a feasible comparator u. Summing over t cancels the intermediate squared distances.

THEOREM — Square-root regret

$$\operatorname{Reg}_T(u) \le \frac{\|\theta_1 - u\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|g_t\|_2^2 \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

For known T and $D, G > 0$, choose $\eta = D/(G\sqrt{T})$:

$$\operatorname{Reg}_T \le DG\sqrt{T}, \qquad \frac{\operatorname{Reg}_T}{T} \le \frac{DG}{\sqrt{T}} \longrightarrow 0.$$

Trade-off: initial distance divided by η, accumulated steps multiplied by η.

The guarantee requires a suitable step size. A fixed positive η independent of T leaves a nonvanishing term $\eta G^2/2$ in the average-regret bound.

OGD with strongly convex losses I

Assumptions: each ℓ_t is α-strongly convex, $\alpha > 0$, with $\|g_t\|_2 \le G$.

THEOREM — A decreasing step gives logarithmic regret

Projected OGD with $\eta_t = \dfrac{1}{\alpha t}$ satisfies

$$\mathbf{Reg}_T(u) \le \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t} \le \frac{G^2}{2\alpha}(1 + \log T) \qquad (u \in \mathcal{K}).$$

The cancellation supplied by curvature

Write $A_t := \|\theta_t - u\|_2^2$. Strong convexity improves the one-step bound:

$$\ell_t(\theta_t) - \ell_t(u) \le \frac{A_t - A_{t+1}}{2\eta_t} - \frac{\alpha}{2} A_t + \frac{\eta_t G^2}{2}.$$

$$\eta_t = \frac{1}{\alpha t} \quad \Longrightarrow \quad \frac{A_t - A_{t+1}}{2\eta_t} - \frac{\alpha}{2} A_t = \frac{\alpha}{2}\big[(t-1)A_t - tA_{t+1}\big].$$

OGD with strongly convex losses II

After summation: distance terms reduce to $-\alpha T A_{T+1}/2 \leq 0$.
Remaining cost:

$$\sum_t \eta_t \frac{G^2}{2},$$

a harmonic sum.

This schedule needs α, but not the horizon T. A global diameter bound is unnecessary here if the stated subgradient bound holds.

What the regret bounds certify

FINITE-TIME PERFORMANCE

A bound B_T controls the cumulative excess loss after any specified T.
Distribution-free: no i.i.d. or stationarity assumption on the loss sequence, under the stated convexity and boundedness conditions.

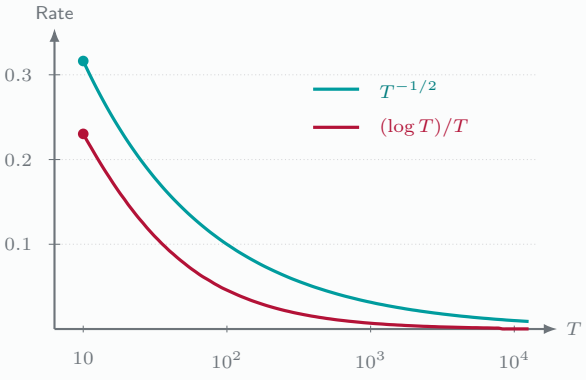
Quantity	Meaning	Role in the bound
D	Diameter of feasible decisions	Larger initial comparator distances
G	Bound on selected subgradients	Larger possible loss changes
α	Curvature of each loss	Harmonic cost with suitable updates
D_R, α_R	Range and curvature of R	Regularization and stability costs in FTRL

Algorithm comparisons: use the same losses, feasible set and comparator.
Scope: the best fixed decision in hindsight.

Tracking a moving optimum: a different benchmark, $\sum_t \ell_t(u_t)$, requiring assumptions on comparator variation.

Average-regret bounds: two decay rates

Rate shapes, with multiplicative constants omitted: $T^{-1/2}$ and $(\log T)/T$.



Logarithmic horizontal axis. Natural logarithm in $\log T$.
Actual bounds include D, G, α and $1 + \log T$. Constants affect finite-horizon comparisons.

Numerical comparison

T	$T^{-1/2}$	$(\log T)/T$
10^2	0.1000	0.04605
10^3	0.0316	0.00691
10^4	0.0100	0.00092

Asymptotic comparison:

$$\frac{(\log T)/T}{T^{-1/2}} = \frac{\log T}{\sqrt{T}} \longrightarrow 0.$$

Strong convexity: a faster decay rate when its assumptions hold.

Further directions: geometry, adaptation and boosting

DEFINITION — Online Mirror Descent (OMD)

With a differentiable strongly convex potential ψ,

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \{\eta_t \langle g_t, u\rangle + D_\psi(u, \theta_t)\}, \quad D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v\rangle.$$

Geometry: D_ψ replaces Euclidean squared distance.
Euclidean: $\psi(u) = \frac{1}{2}\|u\|_2^2$ gives projected OGD.
Negative entropy on the simplex: multiplicative updates.

AdaGrad	Adapts the update geometry using accumulated squared gradients. Useful when coordinates have different scales or frequencies.
AdaBoost	Combines weak classifiers through iterative example reweighting. Connects boosting to multiplicative weights and online learning.

Standard AdaBoost is a boosting procedure over training data. Its iterations need not represent newly arriving observations.

Further reading

Hazan, *Introduction to Online Convex Optimization* (online optimization and mirror descent).
Duchi, Hazan & Singer (2011), *Adaptive Subgradient Methods for Online Learning and Stochastic Optimization*, JMLR.
Freund & Schapire (1997), *A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting*, JCSS.

CHAPTER 5

Regression and inventory

Squared loss, ridge regularization and asymmetric costs

Online regression with squared loss

Available before day t: price, promotions, recent sales and calendar effects.
Encode them in $x_t \in \mathbb{R}^d$. Choose $\theta_t \in \mathcal{K}$ before observing demand y_t.

DEFINITION — Linear prediction and squared error

$$\widehat{y}_t = x_t^\top \theta_t, \qquad \ell_t(\theta) = \frac{1}{2}(x_t^\top \theta - y_t)^2, \qquad e_t := \widehat{y}_t - y_t.$$

Feedback: demand y_t reveals the residual and the loss function.

$$\nabla \ell_t(\theta) = (x_t^\top \theta - y_t) x_t, \qquad g_t = \nabla \ell_t(\theta_t) = e_t x_t.$$

PROPOSITION — Projected OGD update

$$\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t e_t x_t)\,.$$

Before projection: x_t gives the direction, e_t sets its sign and scale.

Predict, observe, correct. The current residual changes the coefficients used for future predictions.

$\mathcal{K} \subseteq \mathbb{R}^d$ is nonempty, closed and convex. An intercept can be included by setting one coordinate of x_t equal to 1.

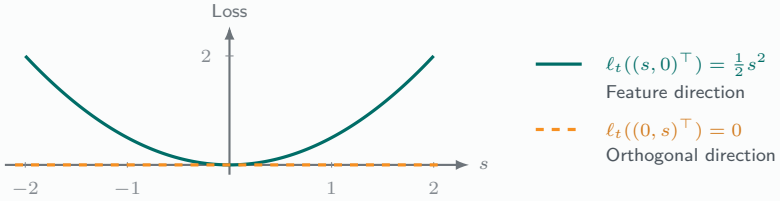
A single observation supplies limited curvature

PROPOSITION — Positive semidefinite Hessian

$$\nabla^2 \ell_t(\theta) = x_t x_t^\top, \qquad v^\top \nabla^2 \ell_t(\theta) v = (x_t^\top v)^2 \geq 0.$$

Convexity: always. **Rank:** at most one.
For $d > 1$, directions orthogonal to x_t have **zero curvature**.

Example: $d = 2$, $x_t = (1, 0)^\top$, $y_t = 0$, so $\ell_t(\theta) = \frac{1}{2}\theta_1^2$.



Strong convexity in $\mathbb{R}^d$ fails for an individual loss when $d > 1$.
Even if $\sum_t x_t x_t^\top \succ 0$, this does not make each loss strongly convex.

Squared regression: the regret guarantee

Bounded setting: $\mathcal{K} = \{\theta : \|\theta\|_2 \le B\}$, $\|x_t\|_2 \le R$, $|y_t| \le Y$, with $B, R > 0$ and $Y \ge 0$.

$$|x_t^\top \theta - y_t| \le RB + Y, \qquad \|\nabla \ell_t(\theta)\|_2 \le \underbrace{(RB+Y)R}_{G_{\mathrm{sq}}}, \qquad D = 2B.$$

THEOREM — A square-root bound for projected OGD

Initialize $\theta_1 \in \mathcal{K}$. For a known horizon T, take $\eta_t = \eta = 2B/(G_{\mathrm{sq}}\sqrt{T})$. Then

$$\mathbf{Reg}_T^{\mathbf{sq}} \le \mathbf{2B(RB+Y)R\sqrt{T}}, \qquad \frac{\mathrm{Reg}_T^{\mathrm{sq}}}{T} \le \frac{2B(RB+Y)R}{\sqrt{T}}.$$

Benchmark: constrained least squares in hindsight

$$u_T^\star \in \arg\min_{u \in \mathcal{K}} \frac{1}{2} \sum_{t=1}^{T} (x_t^\top u - y_t)^2, \qquad \mathrm{Reg}_T^{\mathrm{sq}} := \sum_{t=1}^{T} \big[\ell_t(\theta_t) - \ell_t(u_T^\star)\big].$$

Sequential coefficients versus one fixed vector. The upper bound on average excess loss tends to zero.

No i.i.d. assumption is needed for this regret bound. The comparator uses the entire realized sequence.

Ridge supplies curvature in every direction

DEFINITION — A penalty at every round

For $\lambda > 0$,

$$\widetilde{\ell}_t(\theta) = \underbrace{\frac{1}{2}(x_t^\top \theta - y_t)^2}_{\text{prediction error}} + \underbrace{\frac{\lambda}{2}\|\theta\|_2^2}_{\text{coefficient penalty}} .$$

PROPOSITION — Gradient and uniform curvature

$$\nabla \widetilde{\ell}_t(\theta) = (x_t^\top \theta - y_t)x_t + \lambda\theta, \qquad \nabla^2 \widetilde{\ell}_t(\theta) = x_t x_t^\top + \lambda I_d \succeq \lambda I_d.$$

Every regularized loss is λ-**strongly convex** in Euclidean norm.

Direction ($x_t \neq 0$)	**Squared-loss curvature**	**With ridge**
Parallel to x_t	$\|x_t\|_2^2$	$\|x_t\|_2^2 + \lambda$
Orthogonal to x_t	0	λ

This formula penalizes every coordinate, including an intercept if present. Leaving the intercept unpenalized changes the curvature argument.

Ridge OGD: shrinkage and error correction

Current residual: $e_t = x_t^\top \theta_t - y_t$.

PROPOSITION — The regularized update

$$\theta_{t+1} = \Pi_{\mathcal{K}}\left(\underbrace{(1-\eta_t\lambda)\theta_t}_{\text{ridge shrinkage}} \; \underbrace{-\eta_t e_t x_t}_{\text{data correction}} \right).$$

Mechanism	Effect before projection
Shrinkage	If $0 < \eta_t\lambda < 1$, every coefficient contracts toward zero.
Correction	The residual moves the parameter only in the current feature direction x_t.

EXAMPLE — An unobserved coordinate still receives shrinkage

For $d = 2$ and $x_t = (1, 0)^\top$, the second coordinate receives no data correction:

$$\widetilde{\theta}_{t+1,2} = (1-\eta_t\lambda)\theta_{t,2}.$$

For the later schedule $\eta_t = 1/(\lambda t)$, the shrinkage factor is $1 - 1/t$: it equals 0 at $t = 1$ and lies in $(0, 1)$ for $t > 1$.

The ridge comparator: average loss plus a penalty

DEFINITION — The best fixed regularized coefficient in hindsight

$$u_T^{\text{ridge}} \in \arg\min_{u \in \mathcal{K}} \sum_{t=1}^{T} \widetilde{\ell}_t(u).$$

A penalty at every round accumulates T times

$$\sum_{t=1}^{T} \widetilde{\ell}_t(u) = \frac{1}{2} \sum_{t=1}^{T} (x_t^\top u - y_t)^2 + \frac{T\lambda}{2} \|u\|_2^2.$$

PROPOSITION — Equivalent batch ridge objective

Divide the objective by $T > 0$ without changing its minimizer:

$$u_T^{\text{ridge}} \in \arg\min_{u \in \mathcal{K}} \left\{ \frac{1}{2T} \sum_{t=1}^{T} (x_t^\top u - y_t)^2 + \frac{\lambda}{2} \|u\|_2^2 \right\}.$$

Same minimizer: weight λ for average loss, $T\lambda$ for summed loss.

Ridge OGD: logarithmic regularized regret

Assumptions: $\mathcal{K} = \{\theta : \|\theta\|_2 \le B\}$, $\|x_t\|_2 \le R$, $|y_t| \le Y$ and $\lambda > 0$.

$$\|\nabla \widetilde{\ell}_t(\theta)\|_2 \le \underbrace{(RB + Y)R + \lambda B}_{G_\lambda} \quad (\theta \in \mathcal{K}), \qquad \text{strong convexity parameter: } \lambda.$$

THEOREM — Projected OGD with $\eta_t = 1/(\lambda t)$

Initialize $\theta_1 \in \mathcal{K}$. For every fixed $u \in \mathcal{K}$,

$$\sum_{t=1}^{T} \left[\widetilde{\ell}_t(\theta_t) - \widetilde{\ell}_t(u)\right] \le \frac{G_\lambda^2}{2\lambda}(1 + \log T).$$

In particular,

$$\mathrm{Reg}_T^{\mathrm{ridge}} := \sum_{t=1}^{T} \widetilde{\ell}_t(\theta_t) - \min_{u \in \mathcal{K}} \sum_{t=1}^{T} \widetilde{\ell}_t(u) \le \frac{G_\lambda^2}{2\lambda}(1 + \log T).$$

THE LOSS BEING COMPARED MATTERS

The bound controls **prediction error plus the ridge penalty**.
It does not automatically give logarithmic regret for unpenalized squared error.

Inventory decisions: unequal costs of error

Decision: order $q_t \in \mathcal{K} = [0, Q_{\max}]$ before observing demand $y_t \geq 0$.
Capacity: $Q_{\max} > 0$ represents a storage, budget or supply limit.

DEFINITION — One-period inventory cost

$$\ell_t(q) = \underbrace{c_h(q - y_t)_+}_{\text{excess stock}} + \underbrace{c_b(y_t - q)_+}_{\text{unmet demand}}, \qquad (a)_+ := \max\{a, 0\}.$$

$c_h > 0$: cost per excess unit, such as holding, spoilage or markdown.
$c_b > 0$: cost per unmet unit, such as lost margin or a service penalty.

PROPOSITION — A scaled pinball loss

$$\tau := \frac{c_b}{c_b + c_h}, \qquad \rho_\tau(r) := \tau r_+ + (1 - \tau)(-r)_+,$$

$$\ell_t(q) = (c_b + c_h)\rho_\tau(y_t - q).$$

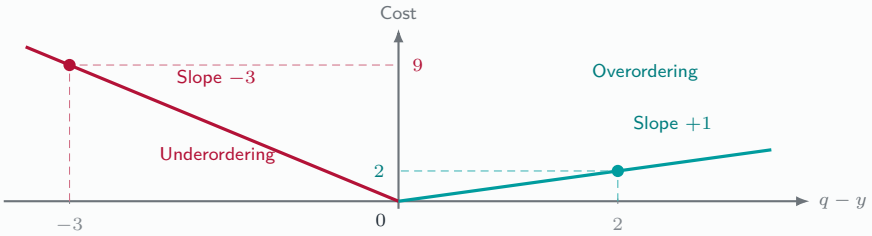
Relative costs determine the quantile level τ.

Simplified model: one-period costs, no stock carryover or lead time. The online protocol below assumes demand is observed after the decision.

When a shortage costs three times as much

Costs per unit: $c_b = 3$, $c_h = 1$. **Critical level:** $\tau = 3/(3+1) = 0.75$.

Order q_t	Demand y_t	$q_t - y_t$	Outcome	Cost
13	16	-3	3 unmet units	$3 \times 3 = 9$
16	14	$+2$	2 excess units	$1 \times 2 = 2$



Target: a 75% demand quantile, subject to capacity.
At equal error magnitude, a shortage costs three times as much as excess stock.

Why the critical quantile minimizes expected cost

Distributional setting: $\mathbb{E}|Y| < \infty$, $F(q) = \mathbb{P}(Y \le q)$ and $F(q^-) = \mathbb{P}(Y < q)$.

$$C(q) := \mathbb{E}[c_h(q - Y)_+ + c_b(Y - q)_+], \qquad \tau = \frac{c_b}{c_b + c_h}.$$

PROPOSITION — The critical quantile, including discrete demand

$$q^\star \in \arg\min_{q \in \mathbb{R}} C(q) \quad \Longleftrightarrow \quad F(q^{\star -}) \le \tau \le F(q^\star).$$

$$\partial C(q) = \Big[(c_b + c_h)F(q^-) - c_b,\ (c_b + c_h)F(q) - c_b\Big], \qquad 0 \in \partial C(q^\star).$$

CONTINUOUS DEMAND AND CAPACITY

If F is continuous and strictly increasing, the unique optimizer is $q^\star = F^{-1}(\tau)$.
On $[0, Q_{\max}]$, an optimal feasible choice is

$$q^\star_{\mathcal{K}} = \min\{Q_{\max}, \max\{0, q^\star\}\}.$$

The clipping rule also applies to any unconstrained optimizer when it is not unique.

With information available before ordering, the analogous target is a conditional demand quantile. The regret analysis below uses a fixed sequence benchmark.

The inventory update is an operational correction

Initialize $q_1 \in [0, Q_{\max}]$. After observing y_t, select $g_t \in \partial \ell_t(q_t)$.

Observed outcome	$\partial \ell_t(q_t)$	Selected g_t	Tentative next order
Shortage: $q_t < y_t$	$\{-c_b\}$	$-c_b$	$q_t + \eta_t c_b$
Exact match: $q_t = y_t$	$[-c_b, c_h]$	0	q_t
Excess: $q_t > y_t$	$\{c_h\}$	c_h	$q_t - \eta_t c_h$

DEFINITION — Projected subgradient step

$$q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t - \eta_t g_t), \qquad \Pi_{[0,Q_{\max}]}(z) = \min\{Q_{\max}, \max\{0, z\}\}.$$

Shortage: raise the next quantity. **Excess:** lower it.
Relative costs: set the asymmetry of these corrections. Projection enforces capacity.

Before clipping, the correction size depends on the cost and step size, not on the error magnitude. The table selects the valid subgradient 0 at a tie.

Inventory OGD: a guarantee against the best fixed quantity

Constants: $D = Q_{\max}$ and $|g_t| \le G := \max\{c_b, c_h\}$.

$$\mathrm{Reg}_T^{\mathrm{inv}} := \sum_{t=1}^{T} \ell_t(q_t) - \min_{q \in [0, Q_{\max}]} \sum_{t=1}^{T} \ell_t(q).$$

THEOREM — Projected OGD with a constant step $\eta > 0$

$$\mathrm{Reg}_T^{\mathrm{inv}} \le \underbrace{\frac{Q_{\max}^2}{2\eta}}_{\text{initial-distance bound}} + \underbrace{\frac{\eta G^2 T}{2}}_{\text{accumulated-step bound}}.$$

For a known horizon T, choose $\eta = Q_{\max}/(G\sqrt{T})$:

$$\mathbf{Reg}_T^{\mathrm{inv}} \le \boldsymbol{Q_{\max} G \sqrt{T}}, \qquad \frac{\mathrm{Reg}_T^{\mathrm{inv}}}{T} \le \frac{Q_{\max} G}{\sqrt{T}} \xrightarrow[T \to \infty]{} 0.$$

WHAT THE GUARANTEE COMPARES

Online: quantities $q_1, \ldots, q_T$ chosen before demand is revealed.
Hindsight: one constant feasible quantity, equivalently a clipped empirical τ-quantile of $y_1, \ldots, y_T$.

No stationarity assumption. The benchmark is one fixed quantity, not a changing conditional quantile.

Chapter recap: the loss determines the decision

Objective	Geometry per round	Best fixed comparator	Regret bound
Squared error	Convex Hessian rank ≤ 1	Constrained least-squares vector	$O(\sqrt{T})$
Ridge	λ-strongly convex	Constrained ridge vector	$O(1 + \log T)$
Inventory cost	Convex, piecewise linear	Clipped empirical τ-quantile	$O(\sqrt{T})$

THREE MODELING CHOICES

Squared loss: learn coefficients that reduce prediction error.
Ridge: penalize large coefficients and supply curvature in every direction.
Asymmetric cost: choose quantities according to the cost of shortages and excess stock.

Common method: a gradient or subgradient correction, followed by projection.
Common benchmark: one fixed decision chosen with hindsight.

Rates are upper bounds under the assumptions and step sizes stated in this chapter. The ridge row concerns regularized losses. A regret guarantee alone does not imply parameter convergence or optimality at every round.